Homework 5 — changed numbers (extra practice)

Modules 5–6 · Homework · No date

The numbers on this sheet are changed from her sheet.

Name _____ Date _____

1. A single 600mg dose of medication was administered orally. The medication is about 90% orally bioavailable and is characterized as having an absorption half-life of approximately 60 minutes, elimination half-life of approximately 5 hours, and apparent volume of distribution of 30 L.

1a Module 5

When does the maximum concentration of drug in the plasma occur?

1b Module 5

What is the expected maximum concentration following this single oral dose?

1c Module 5

What is the expected AUC following this single oral dose?

1d Module 5

If the dose were increased to 1200 mg, when does the maximum concentration of drug in the plasma occur?

1e Module 5

If the dose were increased to 1200 mg, what is the maximum concentration of drug in the plasma that you would expect?

1f Module 5

If the dose were increased to 1200 mg, what is the AUC that you would expect?

2. A 72 kg male patient received multiple IV bolus injections of 12 mg/kg every 6 hours for 36 hours. The medication has an elimination half-life of 3 hours and an apparent

volume of distribution that is 25% of body weight.

2a Module 6

What is the average concentration of drug in the plasma at steady state?

2b Module 6

What is the maximum concentration of drug in the plasma at steady state?

2c Module 6

What is the minimum concentration of drug in the plasma at steady state?

2d Module 6

What is the expected concentration of drug in the plasma 8 hours after administration of the last dose?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working.

1a

Worked by the drill

Answer: 2.9 hr

WHAT THE STEM GIVES

D_0	600mg	*not needed: t_{max} does not depend on the dose*
F	90%	*not needed: F scales height, not timing*
absorption t½	approximately 60 minutes	1 hour, so $k_a = \frac{0.693}{1} = 0.6930\ hr^{-1}$
t½	approximately 5 hours	gives $k = \frac{0.693}{5} = 0.1386\ hr^{-1}$
V_D	30 L	*not needed: t_{max} has no volume in it*

THE WORKING, ONE LINE AT A TIME

SETUP

$$k_a = \frac{0.693}{1\ hr} = 0.6930\ hr^{-1};\ k = \frac{0.693}{5\ hr} = 0.1386\ hr^{-1}$$

The absorption half-life of 60 minutes is 1 hour, so k_a, the absorption rate constant, is $\frac{0.693}{1\ hr} = 0.6930\ hr^{-1}$, and k, the elimination rate constant, is $\frac{0.693}{5\ hr} = 0.1386\ hr^{-1}$. Each is 0.693 over its own half-life because $k\ t_{½} = \ln 2$; both are in reciprocal hours.

ALGEBRA

$$t_{max} = \frac{\ln(0.6930/0.1386)}{(0.6930 - 0.1386)\ hr^{-1}} = \frac{1.6094}{0.5544\ hr^{-1}} = 2.90\ hr$$

t_{max}, the time of the peak, is when the absorption rate equals the elimination rate: $t_{max} = \frac{\ln(k_a/k)}{k_a - k} \cdot \frac{0.6930}{0.1386} = 5.0$, a pure ratio of rate constants, and $\ln 5.0 = 1.6094$; dividing by $(k_a - k) = 0.5544\ hr^{-1}$ inverts the reciprocal hours to hours, 2.90 hr.

> **DOES THE NUMBER MAKE SENSE?**
> Absorption, half-life 1 hour, is five times faster than elimination, half-life 5 hours, so the peak comes a few absorption half-lives in and before one elimination half-life: 2.90 hr, between 1 and 5.

Her homework wording with different numbers, so this is not the graded answer.

1b

Worked by the drill

Answer: 12.04 mg/L

WHAT THE STEM GIVES

D_0	600mg	in the numerator $F\ k_a\ D_0$

F	90%	written 0.9, in the numerator
absorption t½	approximately 60 minutes	gives $k_a = 0.6930\ hr^{-1}$, in the numerator and in $k_a - k$
t½	approximately 5 hours	gives $k = 0.1386\ hr^{-1}$, in $k_a - k$ and $e^{-k\,t_{max}}$
V_D	30 L	in the denominator $V_D(k_a - k)$

THE WORKING, ONE LINE AT A TIME

SETUP

$$k_a = \frac{0.693}{1\ hr} = 0.6930\ hr^{-1};\ k = \frac{0.693}{5\ hr} = 0.1386\ hr^{-1}$$

The absorption half-life of 60 minutes is 1 hour, so k_a, the absorption rate constant, is $\frac{0.693}{1\ hr} = 0.6930\ hr^{-1}$, and k, the elimination rate constant, is $\frac{0.693}{5\ hr} = 0.1386\ hr^{-1}$. Each is 0.693 over its own half-life because $k\ t_{½} = \ln 2$; both are in reciprocal hours.

ALGEBRA

$$C_{max} = \frac{(0.9)(600\ mg)(0.6930\ hr^{-1})}{(30\ L)(0.5544\ hr^{-1})}\left(e^{-(0.1386)(2.90)} - e^{-(0.6930)(2.90)}\right)$$

The single oral dose equation $C_p = \frac{F\,k_a\,D_0}{V_D(k_a - k)}\left(e^{-kt} - e^{-k_a\,t}\right)$ at $t = t_{max} = 2.90$ hr; F = 0.9 is the fraction absorbed, $D_0 = 600$ mg, $V_D = 30$ L. On top, $(0.9)(600)(0.6930) = 374.22\ mg \cdot hr^{-1}$; below, $(30)(0.5544) = 16.632\ L \cdot hr^{-1}$; the reciprocal hours cancel.

ROUND

$$C_{max} = (22.500)(0.5350) = 12.04\ mg/L$$

$\frac{374.22}{16.632} = 22.500$ mg/L is the coefficient. At t_{max} the elimination term $e^{-(0.1386)(2.90)} = 0.6690$ and the absorption term $e^{-(0.6930)(2.90)} = 0.1340$; their difference, 0.5350, times the coefficient gives $C_{max} = 12.04$ mg/L.

> **DOES THE NUMBER MAKE SENSE?**
>
> If the whole absorbed dose were in the body at once the level would be $\frac{0.9 \times 600}{30} = 18$ mg/L; some has been eliminated by 2.90 hr and a little is unabsorbed, so C_{max} is below 18: 12.04 mg/L.

Her homework wording with different numbers, so this is not the graded answer.

1c

Worked by the drill

Answer: 129.9 (mg/L)·hr

WHAT THE STEM GIVES

D_0	600mg	in the numerator F D_0
F	90%	written 0.9, in the numerator
absorption t½	approximately 60 minutes	*not needed: AUC does not depend on k_a*

t½	approximately 5 hours	gives k = 0.1386 hr^{-1}, in the denominator k V_D
V_D	30 L	in the denominator k V_D

THE WORKING, ONE LINE AT A TIME

SETUP

$$AUC = \frac{F\ D_0}{k\ V_D} = \frac{(0.9)(600\ mg)}{(0.1386\ hr^{-1})(30\ L)} = 129.9\ (mg/L)\cdot hr$$

AUC, the area under the concentration–time curve, is the amount reaching the circulation over the clearance, because Cl × AUC = F D_0. Only the absorbed F = 0.9 of the 600 mg counts; Cl = k × V_D = (0.1386 hr^{-1})(30 L) = 4.1580 L/hr, and mg over L/hr is (mg/L)·hr.

> **DOES THE NUMBER MAKE SENSE?**
> AUC is the absorbed dose over clearance; 0.9 × 600 mg over (0.1386 hr^{-1})(30 L) comes to about 130 (mg/L)·hr: 129.9. Doubling the dose would double it.

Her homework wording with different numbers, so this is not the graded answer.

1d Worked by the drill

Answer: 2.9 hr

WHAT THE STEM GIVES

D_0	1200 mg	*not needed: t_{max} has no dose in it*
F	90%	*not needed*
absorption t½	approximately 60 minutes	gives k_a = 0.6930 hr^{-1}
t½	approximately 5 hours	gives k = 0.1386 hr^{-1}
V_D	30 L	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$$t_{max} = \frac{\ln(k_a/k)}{k_a - k} = 2.90\ hr,\ unchanged$$

$t_{max} = \frac{\ln(k_a/k)}{k_a - k}$ contains only k_a = 0.6930 hr^{-1}, the absorption rate constant, and k = 0.1386 hr^{-1}, the elimination rate constant; the dose D_0 does not appear. Doubling the dose to 1200 mg changes neither constant, because both processes are first order, so the peak still comes at 2.90 hr.

> **DOES THE NUMBER MAKE SENSE?**
> Both absorption and elimination are first order, so doubling the dose doubles every concentration but moves no timing: the peak is still at 2.90 hr, the part a value.

Her homework wording with different numbers, so this is not the graded answer.

1e Worked by the drill

Answer: 24.07 mg/L

WHAT THE STEM GIVES

D_0	1200 mg	twice 600 mg, so twice the part b peak
F	90%	unchanged
absorption t½	approximately 60 minutes	unchanged, so t_{max} is unchanged
t½	approximately 5 hours	unchanged
V_D	30 L	unchanged

THE WORKING, ONE LINE AT A TIME

SETUP

$C_{max} = 2 \times 12.04 \text{ mg/L} = 24.07 \text{ mg/L}$

With first-order kinetics the whole curve scales with the dose, so twice the dose gives twice the peak at the same time.

> **DOES THE NUMBER MAKE SENSE?**
> C_{max} is proportional to the dose, so twice 600 mg gives twice 12.04 mg/L: 24.07 mg/L, above the part b peak.

Her homework wording with different numbers, so this is not the graded answer.

1f Worked by the drill

Answer: 259.7 (mg/L)·hr

WHAT THE STEM GIVES

D_0	1200 mg	twice 600 mg, so twice the part c area
F	90%	unchanged
absorption t½	approximately 60 minutes	*not needed: AUC does not depend on k_a*
t½	approximately 5 hours	unchanged
V_D	30 L	unchanged

THE WORKING, ONE LINE AT A TIME

SETUP

$\text{AUC} = 2 \times 129.9 = 259.7 \text{ (mg/L)·hr}$

$\text{AUC} = \frac{F\,D_0}{Cl}$: the amount absorbed over the clearance. F = 0.9 and $Cl = k \times V_D = 4.1580$ L/hr do not change with the dose under first-order kinetics, so AUC is proportional to D_0; 1200 mg is twice 600 mg, and the area doubles from 129.9 to 259.7 (mg/L)·hr.

> **DOES THE NUMBER MAKE SENSE?**

AUC is proportional to the dose at a fixed clearance, so twice 600 mg gives twice 129.9: 259.7 (mg/L)·hr, above the part c area.

Her homework wording with different numbers, so this is not the graded answer.

2a

Worked by the drill

Answer: 34.63 mg/L

WHAT THE STEM GIVES

weight	72 kg	scales both the dose and the volume
D_0	12 mg/kg	scaled to 72 kg: 864 mg, the numerator
τ	every 6 hours	in the denominator V_D k τ
duration	36 hours	*not needed for the number: 12 half-lives, so steady state is reached*
t½	3 hours	gives $k = \frac{0.693}{3} = 0.231\ hr^{-1}$
V_D	25% of body weight	scaled to 72 kg: 18 L, in the denominator

THE WORKING, ONE LINE AT A TIME

UNIT

$$D_0 = (12\ mg/kg)(72\ kg) = 864\ mg;\ V_D = (0.25)(72\ kg) = 18\ L;\ k = \frac{0.693}{3} = 0.231\ hr^{-1}$$

The dose is 12 mg per kilogram and the volume of distribution V_D is 25% of body weight, so both are multiplied by the 72 kg weight: D_0 = 864 mg and V_D = 18 L, the kilograms cancelling. k, the elimination rate constant, is $\frac{0.693}{t½} = \frac{0.693}{3\ hr} = 0.231\ hr^{-1}$.

ALGEBRA

$$C_{avg}^{\infty} = \frac{D_0}{V_D\ k\ \tau} = \frac{864\ mg}{(18\ L)(0.231\ hr^{-1})(6\ hr)} = 34.63\ mg/L$$

C_{avg}^{∞}, the average steady-state concentration, is the dose absorbed per interval over the plasma volume cleared per interval, Cl × τ with Cl = k × V_D: (18 L)(0.231 hr^{-1})(6 hr) = 24.95 L, the hours cancelling. $\frac{864\ mg}{24.95\ L} = 34.63$ mg/L; F, the fraction absorbed, is 1 for an IV dose.

DOES THE NUMBER MAKE SENSE?

$C_0 = \frac{864}{18}$ mg/L per dose and kτ = 0.231 × 6, so the average is C_0 over kτ, a little under three-quarters of C_0: 34.63 mg/L, between the steady-state trough and peak.

Her homework wording with different numbers, so this is not the graded answer.

2b

Worked by the drill

Answer: 64.01 mg/L

WHAT THE STEM GIVES

weight	72 kg	scales both the dose and the volume
D_0	12 mg/kg	scaled to 72 kg: 864 mg, the numerator of C_0
τ	every 6 hours	the exponent $k\tau = 0.231 \times 6 = 1.386$
duration	36 hours	*not needed for the number: 12 half-lives, so steady state is reached*
t½	3 hours	gives $k = \frac{0.693}{3} = 0.231\ hr^{-1}$
V_D	25% of body weight	scaled to 72 kg: 18 L, the denominator of C_0

THE WORKING, ONE LINE AT A TIME

UNIT

$$D_0 = (12\ mg/kg)(72\ kg) = 864\ mg;\ V_D = (0.25)(72\ kg) = 18\ L;\ k = \frac{0.693}{3} = 0.231\ hr^{-1}$$

The dose is 12 mg per kilogram and the volume of distribution V_D is 25% of body weight, so both are multiplied by the 72 kg weight: $D_0 = 864$ mg and $V_D = 18$ L, the kilograms cancelling. k, the elimination rate constant, is $\frac{0.693}{t½} = \frac{0.693}{3\ hr} = 0.231\ hr^{-1}$.

ALGEBRA

$$C_0 = \frac{864\ mg}{18\ L} = 48\ mg/L;\ k\tau = (0.231)(6) = 1.386;\ e^{-k\tau} = 0.2501$$

$C_0 = \frac{D_0}{V_D} = 48$ mg/L is the rise in concentration each dose produces. $k\tau = (0.231\ hr^{-1})(6\ hr) = 1.386$, a pure number, and $e^{-k\tau} = 0.2501$ is the fraction of each dose left when the next is given; the interval $\tau = 6$ hr is 2 half-lives, so a quarter remains.

ALGEBRA

$$C_{max}^{\infty} = \frac{48\ mg/L}{1 - 0.2501} = 64.01\ mg/L$$

Each dose adds $C_0 = 48$ mg/L to what remains of the earlier doses; at steady state the peak is the sum $C_0(1 + e^{-k\tau} + e^{-2k\tau} + \ldots) = \frac{C_0}{1 - e^{-k\tau}}$. $1 - 0.2501 = 0.7499$, so $C_{max}^{\infty} = \frac{48\ mg/L}{0.7499} = 64.01$ mg/L.

> **DOES THE NUMBER MAKE SENSE?**
> Six hours is two half-lives of 3 hours, so a quarter of each dose remains at the next; the peak is $C_0 = 48$ mg/L over $1 - 0.2501$, a third more than 48: 64.01 mg/L, between 48 and 2×48.

Her homework wording with different numbers, so this is not the graded answer.

2c

Worked by the drill

Answer: 16.01 mg/L

WHAT THE STEM GIVES

weight	72 kg	scales the dose and the volume

D_0	12 mg/kg	scaled to 864 mg; C_0 = 48 mg/L, the numerator of the trough line
τ	every 6 hours	the exponent kτ, with $e^{-k\tau} = 0.2501$
duration	36 hours	*not needed for the number: steady state is reached*
t½	3 hours	gives k = 0.231 hr^{-1}; 6 hours is two half-lives
V_D	25% of body weight	scaled to 18 L, the denominator of C_0

THE WORKING, ONE LINE AT A TIME

UNIT

$$D_0 = (12\ \text{mg/kg})(72\ \text{kg}) = 864\ \text{mg};\ V_D = 0.25 \times 72\ \text{kg} = 18\ \text{L};\ C_0 = \frac{864\ \text{mg}}{18\ \text{L}} = 48\ \text{mg/L}$$

The dose and the volume both scale with the 72 kg weight, so C_0, the rise in concentration each dose produces, is the dose over the volume.

SETUP

$$k = \frac{0.693}{3\ \text{hr}} = 0.231\ \text{hr}^{-1};\ k\tau = 0.231 \times 6 = 1.386;\ e^{-k\tau} = 0.2501$$

The printed trough line wants k and τ: k from the 3-hour half-life, and the 6-hour interval gives kτ, a pure number.

ALGEBRA

$$C_{min}^{\infty} = \frac{C_0\, e^{-k\tau}}{1 - e^{-k\tau}} = \frac{(48\ \text{mg/L})(0.2501)}{1 - 0.2501} = 16.01\ \text{mg/L}$$

This is the printed trough line, read directly with C_0 from the stem: no peak is needed. Part b reaches the same number as the 64.01 mg/L peak times $e^{-k\tau}$.

> **DOES THE NUMBER MAKE SENSE?**
> Six hours is two half-lives of 3 hours, so a quarter of each dose is left when the next goes in; the trough is C_0 times 0.2501 over 1 − 0.2501, a third of the 48 mg/L C_0: near 16.01 mg/L, and below C_0.

Her homework wording with different numbers, so this is not the graded answer.

2d

Worked by the drill

Answer: 10.08 mg/L

WHAT THE STEM GIVES

weight	72 kg	scales dose and volume inside the part b peak
D_0	12 mg/kg	inside C_{max}^{∞} = 64.01 mg/L from part b, the level the decay starts from

τ	every 6 hours	*not needed: the time asked is counted from the last dose*
duration	36 hours	*not needed for the number: steady state is reached*
t½	3 hours	gives $k = 0.231\ hr^{-1}$ for the exponent $kt = 0.231 \times 8$
V_D	25% of body weight	inside the part b peak
t	8 hours after administration of the last dose	t in e^{-kt}

THE WORKING, ONE LINE AT A TIME

SETUP

$$C = C_{max}^{\infty}\, e^{-kt} = (64.01\ mg/L)\, e^{-(0.231)(8)} = (64.01)(0.1576) = 10.08\ mg/L$$

After the last dose nothing more is given, so the level follows $C = C_0\, e^{-kt}$ from the steady-state peak $C_{max}^{\infty} = 64.01$ mg/L, the concentration the moment the last dose is in. $kt = (0.231\ hr^{-1})(8\ hr) = 1.848$, a pure number, and $e^{-1.848} = 0.1576$ of the peak remains after 8 hours.

> **DOES THE NUMBER MAKE SENSE?**
> Eight hours is between two and three half-lives of 3 hours, so between an eighth and a quarter of the 64.01 mg/L peak remains: 10.08 mg/L, between 0.125×64.01 and 0.25×64.01.

Her homework wording with different numbers, so this is not the graded answer.